THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 18	There are $2n = 18$ numbers between n^2 and $(n+1)^2$
2	**(a)** 29	$(n+1)^2 - n^2 = 2n + 1 = 29$
3	**(b)** 36	$T_n = n(n+1)/2 = 36$
4	**(d)** 49	$T_n + T_{n+1} = (n+1)^2 = 49$
5	**(d)** 8, 15, 17	$2m = 8$, $m^2 - 1 = 15$, $m^2 + 1 = 17$
6	**(d)** 100	$1^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2 = 100$
7	**(a)** 24	There are $2n = 24$ numbers between n^2 and $(n+1)^2$
8	**(a)** 39	$(n+1)^2 - n^2 = 2n + 1 = 39$
9	**(b)** 55	$T_n = n(n+1)/2 = 55$
10	**(b)** 100	$T_n + T_{n+1} = (n+1)^2 = 100$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 = 100$ (2).	2
12	Take $2m = 6$, $m = 3$ (1); triplet $(6, 8, 10)$ (1).	1+1
13	The first odd number is $n^2 - n + 1 = 7$ (1); $7 + 9 + 11 = 27$ (1).	1+1
14	$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 + 25 = 169$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$T_5 = 15$, $T_6 = 21$ (1+1); sum $= 36 = 6^2$ (1).	1+1+1
16	$22^2 = 484$, $120^2 = 14400$ (1); sum $= 14884$ (1); $122^2 = 14884$ equals the sum, so it is a triplet (1).	1+1+1
17	$1156 = 2^2 \times 17^2$ (1); pairing (1); $\sqrt{1156} = 34$ (1).	1+1+1
	OR: $T_8 = 36$, $T_9 = 45$ (1+1); sum $= 81 = 9^2$ (1).	1+1+1
18	$26^2 = 676$, $168^2 = 28224$ (1); sum $= 28900$ (1); $170^2 = 28900$ equals the sum, so it is a triplet (1).	1+1+1
	OR: $2116 = 2^2 \times 23^2$ (1); pairing (1); $\sqrt{2116} = 46$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$1^3 + 12^3 = 1 + 1728 = 1729$ (2); $9^3 + 10^3 = 729 + 1000 = 1729$ (2); two different ways, so it is a taxicab number (1).	2+2+1
	OR: $13824 = 2^9 \times 3^3$ (1); triplets give $\sqrt[3]{13824} = 24$ (2); check $24^3 = 13824$ (2).	1+2+2
20	$2^3 + 16^3 = 8 + 4096 = 4104$ (2); $9^3 + 15^3 = 729 + 3375 = 4104$ (2); two different ways, so it is a taxicab number (1).	2+2+1
	OR: $17576 = 2^3 \times 13^3$ (1); triplets give $\sqrt[3]{17576} = 26$ (2); check $26^3 = 17576$ (2).	1+2+2

Mistakes to avoid:

- Concluding that a number is a perfect square from its last digit alone: a units digit of 0, 1, 4, 5, 6 or 9 is only *necessary*, not sufficient.
- Pairing primes for a perfect cube: grouping them in **pairs** (square-root rule) instead of **triples**.
- Writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ or $(a+b)^2 = a^2 + b^2$.
- Mixing up 2^3 with 2×3 or 3^2 with 3×2; $4^3 = 64$, not 12.
- Saying the cube root of -8 does not exist; it is -2 because $(-2)^3 = -8$.
- Multiplying or dividing by the wrong prime when making a number a perfect square or cube, instead of looking only at the primes with an incorrect exponent.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 10, 24, 26	$2m = 10$, $m^2 - 1 = 24$, $m^2 + 1 = 26$
2	**(b)** 225	$1^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2 = 225$
3	**(a)** 32	There are $2n = 32$ numbers between n^2 and $(n+1)^2$
4	**(c)** 49	$(n+1)^2 - n^2 = 2n + 1 = 49$
5	**(d)** 78	$T_n = n(n+1)/2 = 78$
6	**(c)** 144	$T_n + T_{n+1} = (n+1)^2 = 144$
7	**(c)** 12, 35, 37	$2m = 12$, $m^2 - 1 = 35$, $m^2 + 1 = 37$
8	**(a)** 441	$1^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2 = 441$
9	**(d)** 40	There are $2n = 40$ numbers between n^2 and $(n+1)^2$
10	**(b)** 61	$(n+1)^2 - n^2 = 2n + 1 = 61$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Take $2m = 16$, $m = 8$ (1); triplet $(16, 63, 65)$ (1).	1+1
12	The first odd number is $n^2 - n + 1 = 13$ (1); $13 + 15 + 17 + 19 = 64$ (1).	1+1
13	$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 + 25 + 27 + 29 = 225$ (2).	2
14	Take $2m = 20$, $m = 10$ (1); triplet $(20, 99, 101)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$T_{10} = 55$, $T_{11} = 66$ (1+1); sum $= 121 = 11^2$ (1).	1+1+1
16	$28^2 = 784$, $195^2 = 38025$ (1); sum $= 38809$ (1); $197^2 = 38809$ equals the sum, so it is a triplet (1).	1+1+1
17	$3136 = 2^6 \times 7^2$ (1); pairing (1); $\sqrt{3136} = 56$ (1).	1+1+1
	OR: $T_{12} = 78$, $T_{13} = 91$ (1+1); sum $= 169 = 13^2$ (1).	1+1+1
18	$30^2 = 900$, $224^2 = 50176$ (1); sum $= 51076$ (1); $226^2 = 51076$ equals the sum, so it is a triplet (1).	1+1+1
	OR: $4356 = 2^2 \times 3^2 \times 11^2$ (1); pairing (1); $\sqrt{4356} = 66$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$2^3 + 24^3 = 8 + 13824 = 13832$ (2); $18^3 + 20^3 = 5832 + 8000 = 13832$ (2); two different ways, so it is a taxicab number (1).	2+2+1
	OR: $21952 = 2^6 \times 7^3$ (1); triplets give $\sqrt[3]{21952} = 28$ (2); check $28^3 = 21952$ (2).	1+2+2
20	$10^3 + 27^3 = 1000 + 19683 = 20683$ (2); $19^3 + 24^3 = 6859 + 13824 = 20683$ (2); two different ways, so it is a taxicab number (1).	2+2+1
	OR: $32768 = 2^{15}$ (1); triplets give $\sqrt[3]{32768} = 32$ (2); check $32^3 = 32768$ (2).	1+2+2

Mistakes to avoid:

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- Pairing primes for a perfect cube: grouping them in **pairs** (square-root rule) instead of **triples**.
- Writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ or $(a+b)^2 = a^2 + b^2$.
- Mixing up 2^3 with 2×3 or 3^2 with 3×2; $4^3 = 64$, not 12.
- Saying the cube root of -8 does not exist; it is -2 because $(-2)^3 = -8$.
- Multiplying or dividing by the wrong prime when making a number a perfect square or cube, instead of looking only at the primes with an incorrect exponent.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 120	$T_n = n(n+1)/2 = 120$
2	**(c)** 225	$T_n + T_{n+1} = (n+1)^2 = 225$
3	**(d)** 14, 48, 50	$2m = 14$, $m^2 - 1 = 48$, $m^2 + 1 = 50$
4	**(d)** 1296	$1^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2 = 1296$
5	**(b)** 50	There are $2n = 50$ numbers between n^2 and $(n+1)^2$
6	**(d)** 99	$(n+1)^2 - n^2 = 2n + 1 = 99$
7	**(b)** 210	$T_n = n(n+1)/2 = 210$
8	**(a)** 324	$T_n + T_{n+1} = (n+1)^2 = 324$
9	**(b)** 18, 80, 82	$2m = 18$, $m^2 - 1 = 80$, $m^2 + 1 = 82$
10	**(c)** 3025	$1^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2 = 3025$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The first odd number is $n^2 - n + 1 = 21$ (1); $21 + 23 + 25 + 27 + 29 = 125$ (1).	1+1
12	$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 + 25 + 27 + 29 + 31 = 256$ (2).	2
13	Take $2m = 24$, $m = 12$ (1); triplet $(24, 143, 145)$ (1).	1+1
14	The first odd number is $n^2 - n + 1 = 31$ (1); $31 + 33 + 35 + 37 + 39 + 41 = 216$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$T_{15} = 120$, $T_{16} = 136$ (1+1); sum $= 256 = 16^2$ (1).	1+1+1
16	$32^2 = 1024$, $255^2 = 65025$ (1); sum $= 66049$ (1); $257^2 = 66049$ equals the sum, so it is a triplet (1).	1+1+1
17	$5625 = 3^2 \times 5^4$ (1); pairing (1); $\sqrt{5625} = 75$ (1).	1+1+1
	OR: $T_{18} = 171$, $T_{19} = 190$ (1+1); sum $= 361 = 19^2$ (1).	1+1+1
18	$34^2 = 1156$, $288^2 = 82944$ (1); sum $= 84100$ (1); $290^2 = 84100$ equals the sum, so it is a triplet (1).	1+1+1
	OR: $7056 = 2^4 \times 3^2 \times 7^2$ (1); pairing (1); $\sqrt{7056} = 84$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$2^3 + 34^3 = 8 + 39304 = 39312$ (2); $15^3 + 33^3 = 3375 + 35937 = 39312$ (2); two different ways, so it is a taxicab number (1).	2+2+1
	OR: $46656 = 2^6 \times 3^6$ (1); triplets give $\sqrt[3]{46656} = 36$ (2); check $36^3 = 46656$ (2).	1+2+2
20	$17^3 + 39^3 = 4913 + 59319 = 64232$ (2); $26^3 + 36^3 = 17576 + 46656 = 64232$ (2); two different ways, so it is a taxicab number (1).	2+2+1
	OR: $59319 = 3^3 \times 13^3$ (1); triplets give $\sqrt[3]{59319} = 39$ (2); check $39^3 = 59319$ (2).	1+2+2

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